TASK 10: ADVANCED SQL QUERY SYSTEM

PROJECT DETAILS

- **Name:** Kaviya S
- **Roll No:** E25AI118
- **Department:** B.Sc Computer Science with AI

SCENARIO

The management team of **PALETTE: Pixel & Pages** wants to perform advanced analysis on customer purchases, product pricing, and sales performance. The management team needs complex reports to identify high-value customers, expensive products, and business trends. This task focuses on implementing Subqueries and Nested Queries to extract deeper insights from the Bookstore Database.

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1. IMPLEMENT SUBQUERIES AND NESTED QUERIES CONCEPT

A **Subquery** is a query written inside another SQL query. It helps solve complex problems by using the result of one query in another query.

Types Covered

- Single-row Subquery
- Multi-row Subquery
- Nested Subquery

Example

1.1 Find Average Price

```
SQL> SELECT AVG(UNIT_PRICE)
  2  FROM BOOK;

AVG(UNIT_PRICE)
---------------
     698.542857
```

1.2 Compare Products with Average Price (Books)

```
SQL> SELECT TITLE,
  2  UNIT_PRICE
  3  FROM BOOK
  4  WHERE UNIT_PRICE >
  5  (
  6  SELECT AVG(UNIT_PRICE)
  7  FROM BOOK
  8  );

TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Designing Data-Intensive Applications
      1250

The Silent Patient
       699

Gone Girl
       799


TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Harry Potter and the Philosopher's Stone
       999

The Hobbit
       850

The Shining
       799


TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Steve Jobs
       899

Sapiens
       899

Percy Jackson and the Lightning Thief
       699


TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
The Da Vinci Code
       799

Angels and Demons
       750

The Inheritance Games
       799
```

```


TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
The Hawthorne Legacy
       850

The Final Gambit
       899

A Good Girls Guide to Murder
       699


TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
The Midnight Library
       799

The Love Hypothesis
       850

Book Lovers
       799
```

---

2. FIND PRODUCTS ABOVE AVERAGE PRICE

2.1 Find products priced above the average price (Books)

```
SQL> SELECT TITLE,
  2  UNIT_PRICE
  3  FROM BOOK
  4  WHERE UNIT_PRICE >
  5  (
  6  SELECT AVG(UNIT_PRICE)
  7  FROM BOOK
  8  );

TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Designing Data-Intensive Applications
      1250

The Silent Patient
       699

Gone Girl
       799


TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Harry Potter and the Philosopher's Stone
       999

The Hobbit
       850

The Shining
       799


TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Steve Jobs
       899

Sapiens
       899

Percy Jackson and the Lightning Thief
       699


TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
The Da Vinci Code
       799

Angels and Demons
       750

The Inheritance Games
       799
```

```
TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
The Hawthorne Legacy
       850

The Final Gambit
       899

A Good Girls Guide to Murder
       699


TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
The Midnight Library
       799

The Love Hypothesis
       850

Book Lovers
       799


18 rows selected.
```

2.2 Display product name, category, and price (Book Title, Book Category, Price)

```
SQL> SELECT B.TITLE,
  2  C.CATEGORY_NAME,
  3  B.UNIT_PRICE
  4  FROM BOOK B
  5  JOIN CATEGORY C
  6  ON B.CATEGORY_ID = C.CATEGORY_ID
  7  WHERE B.UNIT_PRICE >
  8  (
  9  SELECT AVG(UNIT_PRICE)
 10  FROM BOOK
 11  );

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Designing Data-Intensive Applications
Technology
      1250

The Silent Patient
Thriller
       699

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------

Gone Girl
Thriller
       799

Harry Potter and the Philosopher's Stone
Fantasy

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
       999

The Hobbit
Fantasy
       850

The Shining

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Horror
       799

Steve Jobs
Biography
       899
```

```
TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Sapiens
History
       899

Percy Jackson and the Lightning Thief
Young Adult
       699

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------

The Da Vinci Code
Crime
       799

Angels and Demons
Crime

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
       750

The Inheritance Games
Young Adult
       799

The Hawthorne Legacy

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Young Adult
       850

The Final Gambit
Young Adult
       899


TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
A Good Girls Guide to Murder
Young Adult
       699

The Midnight Library
Contemporary Fiction
       799
```

```
TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------

The Love Hypothesis
Modern Romance
       850

Book Lovers
Modern Romance

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
       799


18 rows selected.
```

2.3 Identify expensive products in each category (Books)

```
SQL> SELECT B.TITLE,
  2  C.CATEGORY_NAME,
  3  B.UNIT_PRICE
  4  FROM BOOK B
  5  JOIN CATEGORY C
  6  ON B.CATEGORY_ID = C.CATEGORY_ID
  7  WHERE B.UNIT_PRICE =
  8  (
  9  SELECT MAX(B2.UNIT_PRICE)
 10  FROM BOOK B2
 11  WHERE B2.CATEGORY_ID = B.CATEGORY_ID
 12  );

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
The Great Gatsby
Fiction
       499

Foundation
Sci-Fi
       650

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------

Designing Data-Intensive Applications
Technology
      1250

Gone Girl
Thriller

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
       799

Sherlock Holmes
Mystery
       499

Me Before You

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Romance
       650

Harry Potter and the Philosopher's Stone
Fantasy
       999
```

```
TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
The Shining
Horror
       799

Steve Jobs
Biography
       899

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------

Sapiens
History
       899

Around the World in 80 Days
Adventure

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
       550

The Da Vinci Code
Crime
       799

The Final Gambit

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
Young Adult
       899

To Kill a Mockingbird
Classics
       650


TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------
The Midnight Library
Contemporary Fiction
       799

The Love Hypothesis
Modern Romance
       850
```

```
TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE
----------


16 rows selected.
```

2.4 Find products with price higher than a selected category average (Books)

```
SQL> SELECT B.TITLE,
  2  B.UNIT_PRICE,
  3  C.CATEGORY_NAME
  4  FROM BOOK B
  5  JOIN CATEGORY C
  6  ON B.CATEGORY_ID = C.CATEGORY_ID
  7  WHERE B.UNIT_PRICE >
  8  (
  9  SELECT AVG(B2.UNIT_PRICE)
 10  FROM BOOK B2
 11  WHERE B2.CATEGORY_ID = B.CATEGORY_ID
 12  );

TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------------------
Foundation
       650
Sci-Fi

Gone Girl
       799
Thriller

TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------------------

Me Before You
       650
Romance

Harry Potter and the Philosopher's Stone
       999

TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------------------
Fantasy

The Shining
       799
Horror

Steve Jobs

TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------------------
       899
Biography

Sapiens
       899
History
```

```
--------------------------------------------------------------------------------
UNIT_PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------------------
Around the World in 80 Days
       550
Adventure

The Da Vinci Code
       799
Crime

TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------------------

Angels and Demons
       750
Crime

The Inheritance Games
       799

TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------------------
Young Adult

The Hawthorne Legacy
       850
Young Adult

The Final Gambit

TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------------------
       899
Young Adult

To Kill a Mockingbird
       650
Classics


TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------------------
The Midnight Library
       799
Contemporary Fiction

The Love Hypothesis
       850
Modern Romance
```

```
TITLE
--------------------------------------------------------------------------------
UNIT_PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------------------

Jane Eyre
       620
Classics


17 rows selected.
```

3. IDENTIFY CUSTOMERS WITH MAXIMUM PURCHASES

3.1 Customer who spent the highest amount

```
SQL> SELECT CUSTOMER_ID,
  2  SUM(TOTAL_AMOUNT) AS TOTAL_SPENT
  3  FROM ORDERS
  4  GROUP BY CUSTOMER_ID
  5  HAVING SUM(TOTAL_AMOUNT) =
  6  (
  7  SELECT MAX(TOTAL_SALES)
  8  FROM
  9  (
 10  SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES
 11  FROM ORDERS
 12  GROUP BY CUSTOMER_ID
 13  )
 14  );

CUSTOMER_ID          TOTAL_SPENT
-------------------- -----------
CUST-101                    2348
```

3.2 Customers with maximum number of orders

```
SQL> SELECT CUSTOMER_ID,
  2  COUNT(*) AS NUMBER_OF_ORDERS
  3  FROM ORDERS
  4  GROUP BY CUSTOMER_ID
  5  HAVING COUNT(*) =
  6  (
  7  SELECT MAX(ORDER_COUNT)
  8  FROM
  9  (
 10  SELECT COUNT(*) AS ORDER_COUNT
 11  FROM ORDERS
 12  GROUP BY CUSTOMER_ID
 13  )
 14  );

CUSTOMER_ID          NUMBER_OF_ORDERS
-------------------- ----------------
CUST-101                            2
```

3.3 Customers whose spending is above average

```
SQL> SELECT CUSTOMER_ID,
  2  SUM(TOTAL_AMOUNT) AS TOTAL_SPENT
  3  FROM ORDERS
  4  GROUP BY CUSTOMER_ID
  5  HAVING SUM(TOTAL_AMOUNT) >
  6  (
  7  SELECT AVG(TOTAL_SPENDING)
  8  FROM
  9  (
 10  SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING
 11  FROM ORDERS
 12  GROUP BY CUSTOMER_ID
 13  )
 14  );

CUSTOMER_ID          TOTAL_SPENT
-------------------- -----------
CUST-112                    1799
CUST-101                    2348
CUST-103                    1149
CUST-111                    1999
CUST-117                    1599
```

3.4 Top 5 valuable customers

```
SQL> SELECT *
  2  FROM
  3  (
  4  SELECT CUSTOMER_ID,
  5  SUM(TOTAL_AMOUNT) AS TOTAL_SPENT
  6  FROM ORDERS
  7  GROUP BY CUSTOMER_ID
  8  ORDER BY TOTAL_SPENT DESC
  9  )
 10  WHERE ROWNUM <= 5;

CUSTOMER_ID          TOTAL_SPENT
-------------------- -----------
CUST-101                    2348
CUST-111                    1999
CUST-112                    1799
CUST-117                    1599
CUST-103                    1149
```

---

4. GENERATE COMPLEX BUSINESS QUERIES

Query 1: Best-Selling Product (Book)

- **Display product name (Book Title)**
- **Total quantity sold**
- **Total revenue generated**

```
SQL> SELECT B.TITLE,
  2  SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD,
  3  SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM BOOK B
  5  JOIN ORDER_DETAILS OD
  6  ON B.BOOK_ID = OD.PRODUCT_ID
  7  GROUP BY B.TITLE
  8  HAVING SUM(OD.QUANTITY) =
  9  (
 10  SELECT MAX(TOTAL_QTY)
 11  FROM
 12  (
 13  SELECT SUM(QUANTITY) AS TOTAL_QTY
 14  FROM ORDER_DETAILS
 15  GROUP BY PRODUCT_ID
 16  )
 17  );

TITLE
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_REVENUE
------------------- -------------
The Inheritance Games
                  3          1899
```

Query 2: High-Value Customers

Find customers whose total purchase amount is greater than average customer spending

```
SQL> SELECT CUSTOMER_ID,
  2  SUM(TOTAL_AMOUNT) AS TOTAL_SPENT
  3  FROM ORDERS
  4  GROUP BY CUSTOMER_ID
  5  HAVING SUM(TOTAL_AMOUNT) >
  6  (
  7  SELECT AVG(TOTAL_SPENDING)
  8  FROM
  9  (
 10  SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING
 11  FROM ORDERS
 12  GROUP BY CUSTOMER_ID
 13  )
 14  );

CUSTOMER_ID          TOTAL_SPENT
-------------------- -----------
CUST-112                    1799
CUST-101                    2348
CUST-103                    1149
CUST-111                    1999
CUST-117                    1599
```

Query 3: Category Performance Analysis

Highest revenue-generating category (Book Category)

```
SQL> SELECT *
  2  FROM
  3  (
  4  SELECT C.CATEGORY_NAME,
  5  SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  6  FROM CATEGORY C
  7  JOIN BOOK B
  8  ON C.CATEGORY_ID = B.CATEGORY_ID
  9  JOIN ORDER_DETAILS OD
 10  ON B.BOOK_ID = OD.PRODUCT_ID
 11  GROUP BY C.CATEGORY_NAME
 12  ORDER BY TOTAL_REVENUE DESC
 13  )
 14  WHERE ROWNUM = 1;

CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Young Adult
         1899
```

Number of products sold in each category (Books)

```
SQL> SELECT C.CATEGORY_NAME,
  2  SUM(OD.QUANTITY) AS PRODUCTS_SOLD
  3  FROM CATEGORY C
  4  JOIN BOOK B
  5  ON C.CATEGORY_ID = B.CATEGORY_ID
  6  JOIN ORDER_DETAILS OD
  7  ON B.BOOK_ID = OD.PRODUCT_ID
  8  GROUP BY C.CATEGORY_NAME;

CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
-------------
Fiction
            2

Sci-Fi
            3

Mystery
            1


CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
-------------
Young Adult
            3

Modern Romance
            1

Technology
            1


CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
-------------
Thriller
            2

Romance
            2

Horror
            2


CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
-------------
Contemporary Fiction
            2

Fantasy
            2

Crime
            1


12 rows selected.
```

Average product price by category (Book Categories)

```
SQL> SELECT C.CATEGORY_NAME,
  2  ROUND(AVG(B.UNIT_PRICE),2) AS AVERAGE_PRICE
  3  FROM CATEGORY C
  4  JOIN BOOK B
  5  ON C.CATEGORY_ID = B.CATEGORY_ID
  6  GROUP BY C.CATEGORY_NAME;

CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_PRICE
-------------
Fiction
          499

Sci-Fi
        624.5

Mystery
          499


CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_PRICE
-------------
Biography
          699

Young Adult
        757.5

Modern Romance
        824.5


CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_PRICE
-------------
Technology
         1250

Thriller
          749

Romance
          550


CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_PRICE
-------------
Horror
        674.5

Adventure
        474.5

Contemporary Fiction
        674.5
```

```
CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_PRICE
-------------
Classics
       589.67

Fantasy
        924.5

Crime
          716


CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_PRICE
-------------
History
        674.5


16 rows selected.
```

Query 4: Customer Purchase History

- **Display customer name**
- **Number of orders**
- **Total purchase amount**

```
SQL> SELECT C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  2  COUNT(DISTINCT O.ORDER_ID) AS NUMBER_OF_ORDERS,
  3  SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE_AMOUNT
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.FIRST_NAME,
  8  C.LAST_NAME;

CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_PURCHASE_AMOUNT
-------------------------------------- ---------------- ---------------------
Aarav Sharma                                          1                   699
Priya Nair                                            1                  1149
Arjun Nair                                            1                   450
Ananya Roy                                            2                  2348
Sneha Gupta                                           1                   450
Rohan Verma                                           1                  1999
Vikram Kapoor                                         1                   899
Ishita Menon                                          1                   799
Kabir Singh                                           1                   499
Aditya Mishra                                         1                  1599
Kavya Das                                             1                   750

CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_PURCHASE_AMOUNT
-------------------------------------- ---------------- ---------------------
Rohan Mehta                                           1                   599
Nikhil Bose                                           1                   550
Pooja Reddy                                           1                   799
Anika Iyer                                            1                  1799
Meera Rao                                             1                   650
Neha Varma                                            1                   550
Rahul Joseph                                          1                   399
Diya Patel                                            1                   599

19 rows selected.
```

Most purchased product (Book)

```
SQL> SELECT B.TITLE,
  2  COUNT(*) AS TIMES_PURCHASED
  3  FROM BOOK B
  4  JOIN ORDER_DETAILS OD
  5  ON B.BOOK_ID = OD.PRODUCT_ID
  6  GROUP BY B.TITLE
  7  ORDER BY TIMES_PURCHASED DESC;

TITLE
--------------------------------------------------------------------------------
TIMES_PURCHASED
---------------
The Inheritance Games
              3

The Alchemist
              2

The Great Gatsby
              2


TITLE
--------------------------------------------------------------------------------
TIMES_PURCHASED
---------------
Dune
              2

Sherlock Holmes
              1

Pride and Prejudice
              1


TITLE
--------------------------------------------------------------------------------
TIMES_PURCHASED
---------------
Murder on the Orient Express
              1

The Love Hypothesis
              1

Foundation
              1


TITLE
--------------------------------------------------------------------------------
TIMES_PURCHASED
---------------
Designing Data-Intensive Applications
              1

Me Before You
              1

The Hobbit
              1
```

```
TITLE
--------------------------------------------------------------------------------
TIMES_PURCHASED
---------------
The Silent Patient
              1

Dracula
              1

The Shining
              1


TITLE
--------------------------------------------------------------------------------
TIMES_PURCHASED
---------------
Gone Girl
              1

Harry Potter and the Philosopher's Stone
              1


17 rows selected.
```

5. PREPARE ADVANCED SQL REPORT

Report 1: Premium Product Report (Books)

- **Product Name (Book Title)**
- **Category (Book Category)**
- **Price**
- **Stock Availability**
- **Condition: Products above average price (Books)**

```
SQL> SELECT B.TITLE,
  2  C.CATEGORY_NAME,
  3  B.UNIT_PRICE,
  4  I.STOCK_STATUS
  5  FROM BOOK B
  6  JOIN CATEGORY C
  7  ON B.CATEGORY_ID = C.CATEGORY_ID
  8  JOIN INVENTORY I
  9  ON B.BOOK_ID = I.PRODUCT_ID
 10  WHERE B.UNIT_PRICE >
 11  (
 12  SELECT AVG(UNIT_PRICE)
 13  FROM BOOK
 14  );

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE STOCK_STATUS
---------- --------------------
Designing Data-Intensive Applications
Technology
      1250 Available

The Silent Patient
Thriller
       699 Available

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE STOCK_STATUS
---------- --------------------

Gone Girl
Thriller
       799 Available

Harry Potter and the Philosopher's Stone
Fantasy

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE STOCK_STATUS
---------- --------------------
       999 Available

The Hobbit
Fantasy
       850 Available

The Shining

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE STOCK_STATUS
---------- --------------------
Horror
       799 Available

Steve Jobs
Biography
       899 Available
```

```
TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE STOCK_STATUS
---------- --------------------
Sapiens
History
       899 Available

Percy Jackson and the Lightning Thief
Young Adult
       699 Available

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE STOCK_STATUS
---------- --------------------

The Da Vinci Code
Crime
       799 Available

Angels and Demons
Crime

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE STOCK_STATUS
---------- --------------------
       750 Available

The Inheritance Games
Young Adult
       799 Available

The Hawthorne Legacy

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE STOCK_STATUS
---------- --------------------
Young Adult
       850 Available

The Final Gambit
Young Adult
       899 Available


TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE STOCK_STATUS
---------- --------------------
A Good Girls Guide to Murder
Young Adult
       699 Available

The Midnight Library
Contemporary Fiction
       799 Available
```

```
TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE STOCK_STATUS
---------- --------------------

The Love Hypothesis
Modern Romance
       850 Available

Book Lovers
Modern Romance

TITLE
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
UNIT_PRICE STOCK_STATUS
---------- --------------------
       799 Available


18 rows selected.
```

Report 2: Customer Value Report

- **Customer Name**
- **Total Orders**
- **Total Spending**
- **Customer Category**

```
SQL> SELECT C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  2  COUNT(O.ORDER_ID) AS TOTAL_ORDERS,
  3  SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING,
  4  CASE
  5  WHEN SUM(O.TOTAL_AMOUNT) >= 1500 THEN 'Premium'
  6  WHEN SUM(O.TOTAL_AMOUNT) >= 800 THEN 'Gold'
  7  ELSE 'Regular'
  8  END AS CUSTOMER_CATEGORY
  9  FROM CUSTOMER C
 10  JOIN ORDERS O
 11  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 12  GROUP BY C.FIRST_NAME,
 13  C.LAST_NAME;

CUSTOMER_NAME                          TOTAL_ORDERS TOTAL_SPENDING CUSTOME
-------------------------------------- ------------ -------------- -------
Priya Nair                                        1           1149 Gold
Aarav Sharma                                      1            699 Regular
Arjun Nair                                        1            450 Regular
Ananya Roy                                        2           2348 Premium
Sneha Gupta                                       1            450 Regular
Ishita Menon                                      1            799 Regular
Rohan Verma                                       1           1999 Premium
Vikram Kapoor                                     1            899 Gold
Kabir Singh                                       1            499 Regular
Aditya Mishra                                     1           1599 Premium
Rohan Mehta                                       1            599 Regular

CUSTOMER_NAME                          TOTAL_ORDERS TOTAL_SPENDING CUSTOME
-------------------------------------- ------------ -------------- -------
Nikhil Bose                                       1            550 Regular
Kavya Das                                         1            750 Regular
Meera Rao                                         1            650 Regular
Anika Iyer                                        1           1799 Premium
Pooja Reddy                                       1            799 Regular
Diya Patel                                        1            599 Regular
Rahul Joseph                                      1            399 Regular
Neha Varma                                        1            550 Regular

19 rows selected.
```

Report 3: Sales Performance Report

Product-wise sales (Books)

```
SQL> SELECT B.TITLE,
  2  SUM(OD.QUANTITY) AS QUANTITY_SOLD
  3  FROM BOOK B
  4  JOIN ORDER_DETAILS OD
  5  ON B.BOOK_ID = OD.PRODUCT_ID
  6  GROUP BY B.TITLE;

TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
Gone Girl
            1

Harry Potter and the Philosopher's Stone
            1

Dracula
            1


TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
The Inheritance Games
            3

Dune
            2

The Silent Patient
            1


TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
Sherlock Holmes
            1

Pride and Prejudice
            1

The Alchemist
            2


TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
Murder on the Orient Express
            1

The Love Hypothesis
            1

Foundation
            1
```

```
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
Designing Data-Intensive Applications
            1

The Great Gatsby
            2

Me Before You
            1


TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
The Hobbit
            1

The Shining
            1


17 rows selected.
```

Category-wise revenue (Book Categories)

```
SQL> SELECT C.CATEGORY_NAME,
  2  SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  3  FROM CATEGORY C
  4  JOIN BOOK B
  5  ON C.CATEGORY_ID = B.CATEGORY_ID
  6  JOIN ORDER_DETAILS OD
  7  ON B.BOOK_ID = OD.PRODUCT_ID
  8  GROUP BY C.CATEGORY_NAME;

CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Fiction
          998

Sci-Fi
         1848

Mystery
          499


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Young Adult
         1899

Modern Romance
          899

Technology
         1250


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Thriller
         1498

Romance
         1100

Horror
         1349


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Contemporary Fiction
          949

Fantasy
         1849

Crime
          599


12 rows selected.
```

Top-selling products (Books)

```
SQL> SELECT *
  2  FROM
  3  (
  4  SELECT B.TITLE,
  5  SUM(OD.QUANTITY) AS TOTAL_SOLD
  6  FROM BOOK B
  7  JOIN ORDER_DETAILS OD
  8  ON B.BOOK_ID = OD.PRODUCT_ID
  9  GROUP BY B.TITLE
 10  ORDER BY TOTAL_SOLD DESC
 11  )
 12  WHERE ROWNUM <= 5;

TITLE
--------------------------------------------------------------------------------
TOTAL_SOLD
----------
The Inheritance Games
         3

Dune
         2

The Great Gatsby
         2


TITLE
--------------------------------------------------------------------------------
TOTAL_SOLD
----------
The Alchemist
         2

Gone Girl
         1
```

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High-performing products (Books)

```
SQL> SELECT TITLE
  2  FROM BOOK
  3  WHERE BOOK_ID IN
  4  (
  5  SELECT PRODUCT_ID
  6  FROM ORDER_DETAILS
  7  GROUP BY PRODUCT_ID
  8  HAVING COUNT(*) >= 2
  9  );

TITLE
--------------------------------------------------------------------------------
The Alchemist
Dune
The Inheritance Games
The Great Gatsby
```

Low-performing products (Books)

```
SQL> SELECT B.TITLE
  2  FROM BOOK B
  3  LEFT JOIN ORDER_DETAILS OD
  4  ON B.BOOK_ID = OD.PRODUCT_ID
  5  GROUP BY B.TITLE
  6  HAVING COUNT(OD.PRODUCT_ID) <= 1;

TITLE
--------------------------------------------------------------------------------
Gone Girl
Harry Potter and the Philosopher's Stone
Dracula
Around the World in 80 Days
The Diary of a Young Girl
The Silent Patient
Sherlock Holmes
Pride and Prejudice
Book Lovers
1984
Sapiens

TITLE
--------------------------------------------------------------------------------
Murder on the Orient Express
The Love Hypothesis
Treasure Island
A Good Girls Guide to Murder
The Da Vinci Code
Steve Jobs
The Midnight Library
Foundation
The Final Gambit
The Hawthorne Legacy
Designing Data-Intensive Applications

TITLE
--------------------------------------------------------------------------------
The Fault in Our Stars
Jane Eyre
Me Before You
The Hobbit
The Shining
Percy Jackson and the Lightning Thief
To Kill a Mockingbird
Wings of Fire
Angels and Demons

31 rows selected.
```

High-value customers

```
SQL> SELECT CUSTOMER_ID,
  2  SUM(TOTAL_AMOUNT)
  3  FROM ORDERS
  4  GROUP BY CUSTOMER_ID
  5  HAVING SUM(TOTAL_AMOUNT) > 1500;

CUSTOMER_ID          SUM(TOTAL_AMOUNT)
-------------------- -----------------
CUST-112                          1799
CUST-101                          2348
CUST-111                          1999
CUST-117                          1599
```

Revenue trends

```
SQL> SELECT TO_CHAR(ORDER_DATE,'MON-YYYY') AS SALES_MONTH,
  2  SUM(TOTAL_AMOUNT) AS REVENUE
  3  FROM ORDERS
  4  GROUP BY TO_CHAR(ORDER_DATE,'MON-YYYY');

SALES_MONTH          REVENUE
-------------------- ----------
MAR-2026                17586
```

CONCLUSION

The Advanced SQL Query System successfully uses **Subqueries and Nested Queries** to identify premium products (Books), category performance (Book Categories), customer purchase behavior, high-value customers, and revenue trends. These reports provide deeper business insights and support data-driven decision-making in the bookstore.